

THE ARTHUR-PACKET SUPPORT EQUALITY FOR REAL REDUCTIVE GROUPS

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ABSTRACT. Let ψ be a real Arthur parameter and let ψ_2 be the unipotent parameter in a fixed Jordan decomposition. Adams–Ionov–Mason–Brown–Vogan proved that the microlocal packet of ψ is contained in the support of the two-step Jordan induction of the packet of ψ_2 , and conjectured equality. We prove the reverse inclusion. The argument first passes to a sufficiently positive translate, where the relevant ABV spaces have a common flag-variety model. Perverse dévissage reduces the problem to a singular-support implication, which follows from the incidence calculation underlying the AIMV microstalk comparison. Coherent continuation for individual packet members then returns the result to the original parameter.

1. INTRODUCTION

Let $(G^\Gamma, \mathcal{W})$ be an extended group, let $({}^\vee G^\Gamma, \mathcal{S})$ be a corresponding E -group with second invariant z , and let $\psi: W_{\mathbb{R}} \times SL_2(\mathbb{C}) \rightarrow {}^\vee G^\Gamma$ be an Arthur parameter; then $\psi|_{W_{\mathbb{R}}}$ has bounded image and $\psi|_{\{1\} \times SL_2(\mathbb{C})}$ is algebraic. Write $\psi_{\text{alg}} = \psi|_{\{1\} \times SL_2(\mathbb{C})}$. Its Arthur $\mathfrak{sl}_2$ -triple is

$$e = d\psi_{\text{alg}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f_A = d\psi_{\text{alg}} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = d\psi_{\text{alg}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Fix a Jordan decomposition of ψ in the sense of [AIMV26, Definition 4.1.1], with Levi groups and parabolics $M_2 \subset M_1 \subset G$, $Q_2 = M_2 U_2 \subset M_1$, and $Q_1 = M_1 U_1 \subset G$. On the dual side, ${}^\vee Q_2 = {}^\vee M_2 {}^\vee U_2 \subset {}^\vee M_1$. The parameter viewed in ${}^\vee M_2^\Gamma$ is the unipotent parameter ψ_2 , and the two-step Jordan operator is $R_{q_1}^g R_{q_2}^{m_1}$, where the right- and left-hand operations are normalized cohomological and real parabolic induction, respectively.

For a parabolic $Q = LU$, let $2\rho_u$ be the determinant character of L on $\mathfrak{u}$, and let $(2\rho_u)^\vee: \mathbb{C}^\times \rightarrow {}^\vee L$ be the corresponding algebraic cocharacter on the dual side. Then $z(\rho_u) := (2\rho_u)^\vee(-1) \in Z({}^\vee L)^{\theta_z}$ is the central element of order dividing two [AIMV26, (2.5.1)–(2.5.2)]. Put $z_1 := zz(\rho_{u_1})$ and $z_2 := z_1 z(\rho_{u_2}) = zz(\rho_{u_1})z(\rho_{u_2})$. The extended-group induction maps of [AIMV26, (2.6.1)] have projective types

$$(1.1) \quad \begin{aligned} R_{q_2}^{m_1}: K\Pi^{z_2}(M_2^\Gamma) &\rightarrow K\Pi^{z_1}(M_1^\Gamma), \\ R_{q_1}^g: K\Pi^{z_1}(M_1^\Gamma) &\rightarrow K\Pi^z(G^\Gamma). \end{aligned}$$

For a virtual representation $V = \sum_\pi n_\pi \pi$ of finite length in the Grothendieck group, put $[V] = \{\pi : n_\pi \neq 0\}$, and take unions for sets of virtual representations. Adams–Ionov–Mason–Brown–Vogan prove the inclusion [AIMV26, Theorem 4.6.1(iii)]

$$(1.2) \quad \Pi^z(G^\Gamma)_\psi^{\text{mic}} \subseteq \left[R_{q_1}^g R_{q_2}^{m_1} \Pi^{zz(\rho_{u_1})z(\rho_{u_2})}(M_2^\Gamma)_{\psi_2}^{\text{mic}} \right]$$

and formulate equality as [AIMV26, Conjecture 4.6.5]. The reverse inclusion is not formal because signed contributions from distinct source members may cancel.

Theorem 1.1. *Let $(G^\Gamma, \mathcal{W})$ and $({}^\vee G^\Gamma, \mathcal{S})$ be as above, with second invariant z . Let ψ be a real Arthur parameter and fix a Jordan decomposition in the sense of [AIMV26, Definition 4.1.1]. Set $z_1 = zz(\rho_{u_1})$ and $z_2 = z_1 z(\rho_{u_2})$. Then*

$$(1.3) \quad \Pi^z(G^\Gamma)_\psi^{\text{mic}} = \left[R_{q_1}^g R_{q_2}^{m_1} \Pi^{z_2}(M_2^\Gamma)_{\psi_2}^{\text{mic}} \right].$$

Here is the main geometric step. After a sufficiently positive translation, the ABV components corresponding to $(\mathcal{O}_f, Y)$ and $(\mathcal{O}_g, Y)$ have the common correspondence of flag varieties $X_g \xleftarrow{q_g} Z \xrightarrow{q_f} X_f$, under which $\text{Fib}_g \circ I(\Lambda_f \rightarrow \Lambda_g) \simeq (q_g)_! q_f^* \circ \text{Fib}_f$. The Arthur covectors are $\nu_f = (x_f, e)$ and $\nu_g = (x_g, e)$. For every $P \in \mathcal{P}_f$ we prove

$$(1.4) \quad \nu_g \in \text{SS}(IP) \implies \nu_f \in \text{SS}(P) \quad (P \in \mathcal{P}_f).$$

If a visible simple perverse sheaf occurs in $[IP]_{K_0}$, perverse dévissage gives $\nu_g \in \text{SS}(IP)$. The incidence calculation underlying [AIMV26, Lemma 4.6.2], together with the three-parabolics theorem, then yields (1.4). A coherent-family identity for each source member carries the result back from the good-range translate.

2. BACKGROUND FROM ABV AND AIMV

2.1. ABV packets and Vogan duality. The subscripts f and g refer to the original endpoint and a sufficiently positive translate, respectively. Write $\psi_f = \psi$ and $\psi_g = \psi^N$, where ψ^N is constructed in the next subsection. Let $\mathcal{O}_a \subset {}^\vee \mathfrak{g}$ be the corresponding orbit of infinitesimal character, and let Y denote the geometric ${}^\vee G$ -orbit. The corresponding blocks at the two endpoints have the same Y by Lemma 2.2. Let $\mathcal{D}_a = D^b C^{\vee G^{\text{alg}}} X({}^\vee G^\Gamma, \mathcal{O}_a, Y)$ and $\mathcal{P}_a = P^{\vee G^{\text{alg}}} X({}^\vee G^\Gamma, \mathcal{O}_a, Y)$ be the bounded equivariant constructible derived category on this component and its perverse heart respectively. The corresponding abelian category of equivariant perverse sheaves is that of [ABV92, Definition 7.7]. Let Ξ_a be the set of equivalence classes of complete geometric parameters. A class $\xi \in \Xi_a$ is represented by a ${}^\vee G$ -orbit S_ξ and an irreducible equivariant local system V_ξ on it [ABV92, Definition 7.6]. Write $P_a(\xi)$ for its perverse intermediate extension and $\pi_a(\xi)$ for the corresponding irreducible canonical projective representation of type z [ABV92, (7.10)(d), Theorem 10.4, and Definition 10.8]. The Vogan duality pairing of [ABV92, Theorem 15.12] is diagonal:

$$(2.1) \quad \langle \pi_a(\xi), P_a(\xi') \rangle_a = e(\xi)(-1)^{d(\xi)} \delta_{\xi, \xi'}.$$

Here $e(\xi)$ is the Kottwitz sign and $d(\xi) = \dim_{\mathbb{C}} S_\xi$.

For $C \in \mathcal{D}_a$, its class in the Grothendieck group of $\mathcal{P}_a$ is

$$\begin{aligned} [C]_{K_0} &:= \sum_j (-1)^j [{}^p \mathcal{H}^j(C)] \\ &= \sum_{\xi} \left(\sum_j (-1)^j [{}^p \mathcal{H}^j(C) : P_a(\xi)] \right) [P_a(\xi)]. \end{aligned}$$

Here $[Q : P_a(\xi)]$ denotes the Jordan–Hölder multiplicity of $P_a(\xi)$ in a finite-length perverse sheaf Q .

Let ν_a be the regular Arthur covector. The normalized microstalk functor $\chi_{\psi_a}^{\text{mic}} : \mathcal{P}_a \rightarrow \text{Rep}(A_{\psi_a}^{\text{mic}})$ is exact by [ABV92, Theorem 24.8(c)]. At the regular Arthur covector, the ABV microstalk functor gives the support criterion

$$(2.2) \quad \chi_{\psi_a}^{\text{mic}}(P_a(\xi)) \neq 0 \iff \nu_a \in \text{SS}(P_a(\xi)).$$

This is [ABV92, Theorem 24.8(a),(c), Corollary 24.9(a), and Proposition 19.12(c)]. We call $P_a(\xi)$ *visible at ν_a* when (2.2) holds. By [ABV92, Definition 22.6], the Arthur packet of ψ_a is $\Pi^z(G^\Gamma)_{\psi_a}^{\text{mic}} = \{\pi_a(\xi) : \chi_{\psi_a}^{\text{mic}}(P_a(\xi)) \neq 0\}$.

2.2. Positive translation, projective types, and the good-range bijection. Let ${}^\vee \mathfrak{u}_2 := \text{Lie}({}^\vee U_2)$ be the nilpotent radical of ${}^\vee \mathfrak{q}_2 = \text{Lie}({}^\vee Q_2)$, so that ${}^\vee \mathfrak{q}_2 = {}^\vee \mathfrak{m}_2 \oplus {}^\vee \mathfrak{u}_2$. Relative to the fixed maximal torus ${}^\vee T \subset {}^\vee M_2$, let $\Delta({}^\vee \mathfrak{u}_2)$ be the root set determined by ${}^\vee \mathfrak{u}_2 = \bigoplus_{\alpha \in \Delta({}^\vee \mathfrak{u}_2)} ({}^\vee \mathfrak{m}_1)_\alpha$, and put $\Sigma_2 := \sum_{\alpha \in \Delta({}^\vee \mathfrak{u}_2)} \alpha$. Let $\chi : M_2 \rightarrow \mathbb{C}^\times$ be the algebraic character corresponding to Σ_2 , and let $\chi^\vee : \mathbb{C}^\times \rightarrow Z({}^\vee M_2)$ be the dual algebraic cocharacter. For a sufficiently large positive integer N (see Lemma 2.1), let ψ_2^N be the corresponding central twist of ψ_2 , and let ψ^N be its composite with ${}^\vee M_2^\Gamma \hookrightarrow {}^\vee G^\Gamma$, then we have the formula [AIMV26, (4.6.1)]

$$(2.3) \quad \Pi^{z_2}(M_2^\Gamma)_{\psi_2^N}^{\text{mic}} = \{\sigma \otimes \chi^{2N} : \sigma \in \Pi^{z_2}(M_2^\Gamma)_{\psi_2}^{\text{mic}}\}.$$

Put $\sigma^N = \sigma \otimes \chi^{2N}$. Let $\lambda_{2,f}$ and $\lambda_{2,g}$ be the infinitesimal characters of ψ_2 and ψ_2^N , and let λ_f and λ_g be their images for G^Γ . If $\mu_\chi \in X^*(M_2) \subset X^*(H)$ is the additive weight of χ , the calculation in the [AIMV26, proof of Theorem 4.6.1, immediately after (4.6.1)] gives

$$(2.4) \quad \lambda_{2,g} - \lambda_{2,f} = 2N\mu_\chi, \quad \lambda_g - \lambda_f = 2N\mu_\chi = 4N\rho_{\mathfrak{u}_2}.$$

In the geometric formulas, we regard λ_f and λ_g as elements of $\text{Lie}({}^\vee T) \subset {}^\vee \mathfrak{g}$.

For a semisimple element $\lambda \in {}^\vee \mathfrak{g}$, write ${}^\vee \mathfrak{g}_n(\lambda) = \{X \in {}^\vee \mathfrak{g} : [\lambda, X] = nX\}$, $\mathfrak{n}(\lambda) = \bigoplus_{n \in \mathbb{Z}_{>0}} {}^\vee \mathfrak{g}_n(\lambda)$. Following [AIMV26, Section 3.1], the canonical flat through λ is the affine subspace $\mathcal{F}(\lambda) = \lambda + \mathfrak{n}(\lambda) \subset {}^\vee \mathfrak{g}$. For the two endpoints, define

$$(2.5) \quad \Lambda_f := \mathcal{F}(\lambda_f), \quad \Lambda_g := \mathcal{F}(\lambda_g).$$

Following [AIMV26, Definition 4.1.2], an infinitesimal character $\lambda \in {}^\vee \mathfrak{m}_2$ is in good range for ${}^\vee Q_2 = {}^\vee M_2 {}^\vee U_2$ if

$$(2.6) \quad \operatorname{Re} \alpha(\lambda) > 0, \quad \text{for every } \alpha \in \Delta({}^\vee \mathfrak{u}_2).$$

Let $H \subset M_2$ be the maximal torus dual to the fixed torus ${}^\vee T \subset {}^\vee M_2$, and let $\mathfrak{h} = \operatorname{Lie}(H)$. Set $\kappa := 2\mu_\chi = 4\rho_{\mathfrak{u}_2} \in X^*(H)$, $\lambda_g = \lambda_f + N\kappa$. Here $2\rho_{\mathfrak{u}_2}$ is the determinant weight corresponding to the sum Σ_2 of the roots of ${}^\vee \mathfrak{u}_2$.

Lemma 2.1. *The positive integer N may be chosen so that (2.6) holds for $\lambda = \lambda_g$ and there is a Levi subgroup $L_N \subset G$ containing H and a parabolic subalgebra $\mathfrak{q}_N = \mathfrak{l}_N \oplus \mathfrak{u}_N$, $\mathfrak{l}_N = \operatorname{Lie}(L_N)$ with the following properties:*

- (1) λ_g is in the good range for $\mathfrak{q}_N$ in the sense of [AIMV26, Definition 2.5.3];
- (2) $\lambda_g - \lambda_f = N\kappa$ belongs to $X^*(L_N)$.

Proof. For $\alpha \in \Delta(\mathfrak{g}, \mathfrak{h})$, put $a_\alpha = \operatorname{Re} \langle \lambda_f, \alpha^\vee \rangle$ and $b_\alpha = \langle \kappa, \alpha^\vee \rangle \in \mathbb{Z}$. By [AIMV26, Lemma 2.5.4 and the proof of Theorem 4.6.1, before (4.6.2)], condition (2.6) holds for all sufficiently large N . Since there are only finitely many roots, we may at the same time avoid the finitely many values of N for which $a_\alpha + Nb_\alpha = 0$ for at least one root α with $b_\alpha \neq 0$. Thus

$$(2.7) \quad a_\alpha + Nb_\alpha \neq 0 \quad \text{whenever } b_\alpha \neq 0.$$

Define

$$\mathfrak{l}_N = \mathfrak{h} \oplus \bigoplus_{a_\alpha + Nb_\alpha = 0} \mathfrak{g}_\alpha, \quad \mathfrak{u}_N = \bigoplus_{a_\alpha + Nb_\alpha > 0} \mathfrak{g}_\alpha, \quad \mathfrak{q}_N = \mathfrak{l}_N \oplus \mathfrak{u}_N.$$

Choose a positively definite $W(G, H)$ -invariant form on $X^*(H)_\mathbb{R}$ and use it to identify $X^*(H)_\mathbb{R}$ with $X_*(H)_\mathbb{R}$. Since $\alpha^\vee = 2\alpha/(\alpha, \alpha)$ under this identification, the signs of $\operatorname{Re} \langle \lambda_g, \alpha^\vee \rangle$ give the usual parabolic root decomposition. The roots for which $a_\alpha + Nb_\alpha = 0$ form the Levi subsystem annihilated by $\operatorname{Re} \lambda_g$. The zero and positive root spaces therefore define a parabolic subalgebra $\mathfrak{q}_N$ with Levi factor $\mathfrak{l}_N$. The same root subsystem determines a connected algebraic Levi subgroup $L_N \subset G$ containing H , with Lie algebra $\mathfrak{l}_N$. For every root α of $\mathfrak{u}_N$, one has $\operatorname{Re} \langle \lambda_g, \alpha^\vee \rangle = a_\alpha + Nb_\alpha > 0$, so λ_g is in the good range of [AIMV26, Definition 2.5.3].

If α is a root of $\mathfrak{l}_N$, then $a_\alpha + Nb_\alpha = 0$, and (2.7) forces $b_\alpha = 0$. Thus the integral weight κ vanishes on every coroot of the derived group of L_N . Since L_N is connected with maximal torus H , restriction to H identifies $X^*(L_N)$ with the weights $\nu \in X^*(H)$ satisfying $\langle \nu, \alpha^\vee \rangle = 0$ for every root α of L_N . It follows that $\kappa \in X^*(L_N)$, and hence $\lambda_g - \lambda_f = N\kappa \in X^*(L_N)$. $\square$

Fix N satisfying the lemma and define $T = T_{L_N}(\lambda_g \rightarrow \lambda_f)$ to be the coherent continuation operator. It maps $K\Pi_{\lambda_g}^z(G^\Gamma)$ to $K\Pi_{\lambda_f}^z(G^\Gamma)$, and the coherent continuations on the individual $(\mathfrak{g}, K_\gamma)$ -module Grothendieck groups, defined in [AIMV26, Definition 2.10.3], are summed over strong real forms; see (5.1)–(5.2).

Write $R = R_{\mathfrak{q}_1}^{\mathfrak{g}} R_{\mathfrak{q}_2}^{\mathfrak{m}_1}$, $\tilde{R} = e(G^\Gamma) R_{\mathfrak{q}_1}^{\mathfrak{g}} R_{\mathfrak{q}_2}^{\mathfrak{m}_1} e(M_2^\Gamma)$. Here $e(G^\Gamma)$ and $e(M_2^\Gamma)$ are the Kottwitz signs of [AIMV26, Theorem 4.2.3]. Define the restrictions of R and $\tilde{R}$ by

$$\begin{aligned} R_N, \tilde{R}_N &: K\Pi_{\lambda_{2,g}}^z(M_2^\Gamma) \longrightarrow K\Pi_{\lambda_g}^z(G^\Gamma), \\ R_0, \tilde{R}_0 &: K\Pi_{\lambda_{2,f}}^z(M_2^\Gamma) \longrightarrow K\Pi_{\lambda_f}^z(G^\Gamma). \end{aligned}$$

For the translated parameters ψ_2^N and ψ^N , [AIMV26, Remark 4.5.2] gives a bijection $R_N : \Pi^{z^2}(M_2^\Gamma)_{\psi_2^N}^{\operatorname{mic}} \longrightarrow \Pi^z(G^\Gamma)_{\psi^N}^{\operatorname{mic}}$. This map sends each irreducible, up to sign, to a single nonzero irreducible. Thus, for every σ^N there are unique $\tau_g \in \Pi^z(G^\Gamma)_{\psi^N}^{\operatorname{mic}}$ and signs $\beta_{\sigma,N}, \alpha_{\sigma,N} \in \{\pm 1\}$ such that

$$(2.8) \quad R_N \sigma^N = \beta_{\sigma,N} \tau_g, \quad \tilde{R}_N \sigma^N = \alpha_{\sigma,N} \tau_g.$$

The second equality follows because the Kottwitz operators are diagonal.

2.3. The flag-variety model.

Lemma 2.2. *The two endpoints have the same geometric Y -coordinate. If ${}^\vee R_a = Z_{\vee G}(\exp(2\pi i \lambda_a))^\circ$, then ${}^\vee R_f = {}^\vee R_g =: {}^\vee R$, where $(\cdot)^\circ$ denotes the identity component. Their symmetric subgroup is also the same: ${}^\vee K = Z_{\vee R}(y)$; set ${}^\vee \theta = \operatorname{Ad}(y)|_{{}^\vee R}$. For $a \in \{f, g\}$, let ${}^\vee P_a = \operatorname{Stab}_{{}^\vee R}(\Lambda_a)$, let ${}^\vee \mathfrak{u}_a$ be the nilradical of $\operatorname{Lie}({}^\vee P_a)$, and put $X_a = {}^\vee R / {}^\vee P_a$. The normalized fiber restriction of [AIMV26, (3.1.2)] gives an equivalence Fib_a from $\mathcal{D}_a$ to $D^b C^{\vee K^{\operatorname{alg}}}(X_a)$. Its restriction to the perverse hearts carries the Arthur covector to $\nu_a = (x_a, e)$, where $x_a = 1^\vee P_a$, and carries the normalized Arthur microstalk to the microstalk at ν_a . Put $Z = {}^\vee R / ({}^\vee P_g \cap {}^\vee P_f)$, and let $q_a : Z \rightarrow X_a$ be the natural projections. Under these equivalences, $I(\Lambda_f \rightarrow \Lambda_g)$ becomes $I_{\operatorname{fib}} = (q_g)_! q_f^*$.*

Proof. Let $\varphi_a = \varphi_{\psi_a}$ be the Langlands parameter associated with ψ_a , for $a \in \{f, g\}$. By [AIMV26, Proposition 3.1.2], the first coordinate of the corresponding geometric parameter is $y_a = \exp(\pi i \lambda_a) \varphi_a(j)$. The positive twist defining ψ_g from ψ_f is trivial on j and on the Arthur $SL_2(\mathbb{C})$; see the proof of [AIMV26, Theorem 4.6.1, before (4.6.1)]. Hence $\varphi_g(j) = \varphi_f(j)$, and the Arthur nilpotent element e is the same at both endpoints.

The duality of the fixed tori identifies the weight μ_χ with $d\chi^\vee(1) \in \text{Lie}(\vee T)$. Thus, in the geometric realization of (2.4), $\lambda_g - \lambda_f = 2N d\chi^\vee(1)$, and $\exp(\pi i(\lambda_g - \lambda_f)) = \chi^\vee(e^{2\pi i N}) = 1$, while $\exp(2\pi i(\lambda_g - \lambda_f)) = \chi^\vee(e^{4\pi i N}) = 1$. It follows that $y_g = y_f =: y$ and ${}^\vee R_f = {}^\vee R_g =: {}^\vee R$. Consequently the two symmetric subgroups also coincide: ${}^\vee K = Z_{\vee R}(y)$.

For $a \in \{f, g\}$, put $X_a = {}^\vee R / {}^\vee P_a$. The associated-bundle description of [AIMV26, Section 3.1 and (3.1.2)] identifies the corresponding geometric component with ${}^\vee G \times {}^\vee K X_a$. Let $i_a : X_a \hookrightarrow {}^\vee G \times {}^\vee K X_a$ be the fiber over $1^\vee K$, and put $d = \dim_{\mathbb{C}}({}^\vee G / {}^\vee K)$. The normalized restriction $\text{Fib}_a = i_a^*[-d]$ is perverse t -exact. Indeed, every ${}^\vee G$ -orbit in the associated bundle is ${}^\vee G \times {}^\vee K S$ for a ${}^\vee K$ -orbit $S \subset X_a$, and has complex dimension $d + \dim_{\mathbb{C}} S$. On a local analytic section of ${}^\vee G \rightarrow {}^\vee G / {}^\vee K$, the orbit stratification is the product of a smooth d -dimensional base with the ${}^\vee K$ -orbit stratification of X_a ; the shift $[-d]$ is exactly the middle-perversity normalization. Thus Fib_a restricts to an equivalence of perverse hearts. Under this equivalence, (y, Λ_a) corresponds to $x_a = 1^\vee P_a \in X_a$. By [AIMV26, Propositions 3.7.1 and 3.7.3], the Arthur covector and its component group are carried to the fiber covector $\nu_a = (x_a, e)$. Normal slices to ${}^\vee G \times {}^\vee K S$ are the corresponding normal slices to S in the fiber, so the same normalization identifies the Arthur microstalk with the microstalk at ν_a .

Since the two endpoints have the same groups ${}^\vee R$ and ${}^\vee K$, the translation correspondence restricts to the projections $q_a : Z = {}^\vee R / ({}^\vee P_g \cap {}^\vee P_f) \rightarrow X_a$. The definition of convolution in [AIMV26, Section 3.5] and the calculation in the proof of [AIMV26, Lemma 4.6.2] identify the induced functor with $I_{\text{fib}} = (q_g)_! q_f^*$. The normalizing shift d is the same at both endpoints, so the two shifts cancel. Hence $\text{Fib}_g \circ I(\Lambda_f \rightarrow \Lambda_g) \simeq I_{\text{fib}} \circ \text{Fib}_f$. $\square$

2.4. Continuation–convolution duality. Let $I_K = K_0(I)$ be the homomorphism induced by I on the Grothendieck group. With the canonical flats defined in (2.5), we apply [AIMV26, Theorem 3.6.3] with $L = L_N$, $\lambda' = \lambda_g$, $\lambda = \lambda_f$, $\Lambda = \Lambda_f$, and $\Lambda' = \Lambda_g$. The two hypotheses (GR1) and (GR2) are Lemma 2.1(1) and (2), respectively. The theorem therefore identifies $T_{L_N}(\lambda_g \rightarrow \lambda_f)$ as the dual transpose of the globally normalized convolution $I(\Lambda_f \rightarrow \Lambda_g)$ under the Vogan duality pairing:

$$(2.9) \quad \langle Tv, x \rangle_f = \langle v, I_K x \rangle_g.$$

We use the sign convention of [AIMV26, Theorem 3.6.3] in (2.9).

2.5. The three-parabolics identity. We shall use the following specialization of the three-parabolics theorem [AIMV26, Theorem A.2.1].

Lemma 2.3. *For the translation datum (Λ_f, Λ_g) constructed above, with $\lambda_g - \lambda_f = 4N\rho_{u_2}$ and $N \gg 0$, the set of $r \in {}^\vee P_f {}^\vee P_g$ satisfying $\text{Ad}(r)e \in {}^\vee \mathfrak{u}_f$ is exactly ${}^\vee P_f$.*

Proof. Let ${}^\vee Q_2 = {}^\vee M_2 {}^\vee U_2 \subset {}^\vee M_1$ be the dual Jordan parabolic and put ${}^\vee Q = {}^\vee Q_2 \cap {}^\vee R$. By [AIMV26, Lemma 4.3.1], the pseudo-Levi of ${}^\vee M_1$ determined by λ_f is ${}^\vee R$, and [AIMV26, Lemma 4.4.1(i)] shows that ${}^\vee Q$ is a parabolic subgroup of ${}^\vee R$ with Levi factor ${}^\vee L := Z_{{}^\vee M_2}(\exp(2\pi i \lambda_f))^\circ$; write ${}^\vee Q = {}^\vee L {}^\vee U$.

For $a \in \{f, g\}$, write $\lambda_a = \frac{1}{2}h + \zeta_a$, and put $\zeta_a^{\mathbb{R}} = \text{Re } \zeta_a$ and $\lambda_a^{\mathbb{R}} = \frac{1}{2}h + \zeta_a^{\mathbb{R}}$. The Arthur triple (e, f_a, h) lies in ${}^\vee \mathfrak{m}_2$. Since $\lambda_f = \frac{1}{2}h + \zeta_f$ and ζ_f centralizes the Arthur $\mathfrak{sl}_2$, its $\text{ad}(\lambda_f)$ -weights are $1, -1, 0$. Thus the triple is fixed by $\exp(2\pi i \lambda_f)$ and lies in ${}^\vee \mathfrak{r}$, so $(e, f_a, h) \subset {}^\vee \mathfrak{r} \cap {}^\vee \mathfrak{m}_2 = {}^\vee \mathfrak{l}$. After conjugation by ${}^\vee L$, choose ${}^\vee T \subset {}^\vee B \subset {}^\vee R$ so that ${}^\vee Q$ is standard, $h \in \text{Lie}({}^\vee T)$, and h is dominant for the positive roots of ${}^\vee L$.

Replacing λ_a by $\lambda_a^{\mathbb{R}}$ does not change the grading on ${}^\vee \mathfrak{r}$. Indeed, for every root α of ${}^\vee R$, the definition of ${}^\vee R$ gives $\exp(2\pi i \alpha(\lambda_a)) = 1$, hence $\alpha(\lambda_a) \in \mathbb{Z}$. Since $\alpha(h)$ is real, $\alpha(\text{Im } \zeta_a) = 0$. Thus $\text{Im } \zeta_a$ is central in ${}^\vee \mathfrak{r}$. Hence ${}^\vee \mathfrak{r}_n(\lambda_a) = {}^\vee \mathfrak{r}_n(\lambda_a^{\mathbb{R}})$ for every $n \in \mathbb{Z}$. Since the stabilizer ${}^\vee P_a = \text{Stab}_{{}^\vee R}(\Lambda_a)$ has Lie algebra $\text{Lie}({}^\vee P_a) = \bigoplus_{n \geq 0} {}^\vee \mathfrak{r}_n(\lambda_a)$, ${}^\vee P_a$ is also the grading parabolic attached to $\lambda_a^{\mathbb{R}}$.

We verify the hypotheses of [AIMV26, Theorem A.2.1]. First, the preceding argument shows that $\lambda_f^{\mathbb{R}}$ and $\lambda_g^{\mathbb{R}}$ take integral values on every root of ${}^\vee R$. Second, Definition 4.1.1(4) of [AIMV26] identifies ψ_2 as a unipotent Arthur parameter for ${}^\vee M_2^\Gamma$, while [AIMV26, Definition 2.3.3] says that $\psi_2(\mathbb{C}^\times) \subset Z({}^\vee M_2)$. Differentiating gives $\zeta_f \in Z({}^\vee \mathfrak{m}_2)$. By the positive translation formula [AIMV26, (4.6.1)], $\zeta_g - \zeta_f$ is the differential of a central cocharacter of ${}^\vee M_2$. Hence $\zeta_f^{\mathbb{R}}$ and $\zeta_g^{\mathbb{R}}$ are central on ${}^\vee L$, and $\alpha(h) = 2\alpha(\lambda_f^{\mathbb{R}}) \in 2\mathbb{Z}$ for every root α of ${}^\vee L$. The grading defined by h on ${}^\vee \mathfrak{l}$ is therefore even.

For the weakly fair condition, fix $\alpha \in \Delta({}^\vee U, {}^\vee T)$ and $0 \neq X_\alpha \in {}^\vee \mathfrak{t}_\alpha$. Let $\zeta_f^{0,1}$ denote the antiholomorphic part of the differential of $\psi|_{\mathbb{C}^\times}$. On $\mathbb{C}X_\alpha$, this restriction has exponents $A = \alpha(\zeta_f)$ and $B = \alpha(\zeta_f^{0,1})$. Single-valuedness gives $A - B \in \mathbb{Z}$, and boundedness gives $\operatorname{Re}(A + B) = 0$. The preceding argument gives $A \in \mathbb{R}$, whence $B \in \mathbb{R}$ and $B = -A$. The S^1 -weight of the root line is therefore $m_\alpha = A - B = 2A = 2\alpha(\zeta_f^\mathbb{R})$. By definition, ${}^\vee \mathfrak{u}_2$ is the sum of the positive S^1 -weight spaces. Since ${}^\vee \mathfrak{u} \subset {}^\vee \mathfrak{u}_2$, we have $m_\alpha > 0$, and hence $\alpha(\zeta_f^\mathbb{R}) > 0$.

Finally, N was chosen so that $\operatorname{Re} \alpha(\lambda_g) > 0$ for every root α of ${}^\vee \mathfrak{u}_2$. Since ${}^\vee \mathfrak{u} \subset {}^\vee \mathfrak{u}_2$ and $\lambda_g^\mathbb{R}$ gives the same grading on ${}^\vee \mathfrak{t}$, every $\lambda_g^\mathbb{R}$ -weight on ${}^\vee U$ is strictly positive. All the hypotheses of [AIMV26, Theorem A.2.1] now hold for ${}^\vee R$, the parabolic ${}^\vee Q = {}^\vee L {}^\vee U$, the Arthur triple (e, f_A, h) , and the grading elements $\lambda_f^\mathbb{R}, \lambda_g^\mathbb{R}$. Its conclusion is the asserted identity. $\square$

3. MICROSUPPORT PRELIMINARIES

3.1. Cotangent conventions. Kashiwara–Schapira microsupport is a subset of the real cotangent bundle of the underlying real analytic manifold. For a complex manifold X , we use the convention of [KS85, Section 8.5.1], identifying $T_x^{*(1,0)} X$ with $T_{x,\mathbb{R}}^* X$ by $\xi \mapsto 2 \operatorname{Re} \xi$. The factor 2 makes the canonical real one-form twice the real part of the holomorphic canonical one-form. By [KS85, Theorem 8.5.2], the microsupport of a complex constructible sheaf is invariant under the natural $\mathbb{C}^\times$ -action.

3.2. Canonical relations. Throughout, $\operatorname{Supp}(F)$ denotes the closed support of a complex F . Thus, for a locally closed immersion $j : U \hookrightarrow X$, one has $\operatorname{Supp}(j_! F) \subseteq \overline{j(\operatorname{Supp}(F))}$.

Let $a : Y \rightarrow X$ be a C^1 map of real analytic manifolds. At $y \in Y$, the transpose of $da_y : T_y Y \rightarrow T_{a(y)} X$ is ${}^t da_y : T_{a(y)}^* X \rightarrow T_y^* Y$, characterized by $({}^t da_y \xi)(v) = \xi(da_y(v))$ for $v \in T_y Y$. On $Y \times_X T^* X$, define $a_d(y, \xi) = (y, {}^t da_y \xi) \in T^* Y$ and $a_\pi(y, \xi) = (a(y), \xi) \in T^* X$. We use the following results of Kashiwara–Schapira.

Proposition 3.1. *Let $F \in D_c^b(Y)$ and $G \in D_c^b(X)$.*

(i) *If a is proper on $\operatorname{Supp}(F)$, then*

$$(3.1) \quad \operatorname{SS}(Ra_! F) \subseteq a_\pi(a_d^{-1} \operatorname{SS}(F)).$$

If a is a closed immersion, equality holds.

(ii) *If a is a smooth submersion, then*

$$(3.2) \quad \operatorname{SS}(a^{-1} G) = a_d(a_\pi^{-1} \operatorname{SS}(G)).$$

Proof. For (i), properness on $\operatorname{Supp}(F)$ identifies $Ra_! F$ with $Ra_* F$, so the assertion, including equality for a closed immersion, is [KS85, Proposition 4.1.1(i),(ii)]. Assertion (ii) is [KS85, Proposition 4.1.2(i)]. Since when a is a submersion, da_y is surjective and hence its transpose ${}^t da_y$, and therefore a_d , is injective. $\square$

For an open immersion $j : V \hookrightarrow X$, (3.2) gives the locality formula

$$(3.3) \quad \operatorname{SS}(j^{-1} F) = \operatorname{SS}(F) \cap T^* V.$$

3.3. Perverse dévissage at a regular conormal point. Let the pro-algebraic group H act with finitely many orbits on the smooth complex variety X , and fix a regular conormal datum $\nu = (x, \xi)$ over an H -orbit S of complex dimension s . Choose the normal slice and Morse pair $J_\nu \supset K_\nu$ of [ABV92, (24.10)(a)]. For every $C \in D_{c,H}^b(X)$, define the normalized Morse complex by

$$(3.4) \quad \mu_\nu(C) := R\Gamma(J_\nu, K_\nu; C)[-s] \in D^b(\operatorname{Vect}_\mathbb{C}).$$

Here $R\Gamma(J_\nu, K_\nu; C)$ is the relative hypercohomology complex of C on the Morse pair; explicitly, it is $\operatorname{Cone}(R\Gamma(J_\nu; C) \rightarrow R\Gamma(K_\nu; C))[-1]$. This is the construction of [ABV92, Definition 24.11] for bounded constructible complexes. As ν varies in a connected component of the regular conormal locus, the groups $H^i \mu_\nu(C)$ form a local system. In particular, different choices of normal slices and Morse pairs give locally canonically isomorphic Morse groups. For a perverse sheaf P , [ABV92, (24.10)(b),(c) and Theorem 24.8(c)] gives

$$(3.5) \quad H^r \mu_\nu(P) = 0 \quad (r \neq 0), \quad H^0 \mu_\nu(P) \simeq \operatorname{For} \chi_\nu^{\operatorname{mic}}(P).$$

Here $\operatorname{For} : \operatorname{Rep}(A_\nu^{\operatorname{mic}}) \rightarrow \operatorname{Vect}_\mathbb{C}$ in (3.5) is the forgetful functor. Let $H_\nu = \operatorname{Stab}_H(x, \xi)$ and $A_\nu^{\operatorname{mic}} = H_\nu / (H_\nu)_0$, where $(H_\nu)_0$ is the identity component. This is the profinite equivariant micro-fundamental group at ν [ABV92, Definition 24.7]. We write $\operatorname{Rep}(A_\nu^{\operatorname{mic}})$ for the category of finite-dimensional continuous complex representations.

The H -equivariant structure on P makes $H^0\mu_\nu(P)$ an object of this category, denoted by $\chi_\nu^{\text{mic}}(P)$. If $\nu = (x, 0)$ lies over an open orbit, the normal slice is a point and (3.4) is the shifted stalk $i_x^{-1}C[-s]$.

Lemma 3.2. *For every $C \in D_{c,H}^b(X)$ and every $j \in \mathbb{Z}$, there is a natural isomorphism*

$$(3.6) \quad H^j\mu_\nu(C) \simeq \text{For } \chi_\nu^{\text{mic}}({}^p\mathcal{H}^j(C)).$$

Proof. Apply the triangulated functor μ_ν to the finite perverse tower of C . The associated spectral sequence is $E_2^{r,s} = H^r\mu_\nu({}^p\mathcal{H}^s(C)) \implies H^{r+s}\mu_\nu(C)$. By (3.5), only the row $r = 0$ can be nonzero. The spectral sequence therefore degenerates, and for each total degree j the induced filtration has the single nonzero graded piece $H^0\mu_\nu({}^p\mathcal{H}^j(C)) \simeq \text{For } \chi_\nu^{\text{mic}}({}^p\mathcal{H}^j(C))$. The edge morphism gives the natural isomorphism (3.6). $\square$

Lemma 3.3. *For $C \in \mathcal{D}_g$, if $\mu_{\nu_g}(C) \neq 0$, then $\nu_g \in \text{SS}(C)$.*

Proof. Suppose first that $e = 0$, and let S_g be the orbit of x_g . If S_g were not open, finiteness of the orbit decomposition would provide an orbit $S' \neq S_g$ with $x_g \in \overline{S'}$. The closure is H -stable, so $S_g \subset \overline{S'}$. Since the zero section over S' is contained in $T_{S'}^*X$, this would imply

$$(x_g, 0) \in \overline{T_{S'}^*X},$$

contradicting the regularity of $\nu_g = (x_g, 0)$ in the sense of [ABV92, (24.1)]. Hence S_g is open and $\mu_{\nu_g}(C) = i_{x_g}^{-1}C[-s] \neq 0$. Thus $x_g \in \text{Supp}(C)$, equivalently $(x_g, 0) \in \text{SS}(C)$.

Now suppose that $e \neq 0$. Choose the adapted real function f_g used to construct $J_{\nu_g} \supset K_{\nu_g}$ in [ABV92, (24.10)(a)], with $f_g(x_g) = 0$ and $df_g|_{x_g} = 2\text{Re}(e)$, as prescribed by Section 3.1. We may choose its extension off the normal slice so that $f_g|_{S_g}$ has a nondegenerate critical point at x_g . Indeed, in local product coordinates, add a sufficiently small generic real quadratic form q in the tangential variables, chosen so that the Hessian of $(f_g + q)|_{S_g}$ at x_g is nondegenerate. The form q vanishes on the normal slice and has zero value and differential at x_g ; replacing f_g by $f_g + q$ therefore changes neither $df_g|_{x_g}$ nor the normal Morse pair (J_{ν_g}, K_{ν_g}) , thus we still use the notation f_g .

Choose a control-data product neighborhood $U \simeq B \times N$ of x_g , where $B \subset S_g$ is a contractible ball and N is the normal slice. Let $(M_{\text{tan}}^+, M_{\text{tan}}^-)$ be the Morse pair of $f_g|_{S_g}$ at x_g . After shrinking U , the added quadratic form depends only on B , and the normal pair on N remains (J_{ν_g}, K_{ν_g}) . The stratified Morse product theorem [GM88, Theorem 3.7, pp. 65–66, and §6.A.1, pp. 222–223] identifies the local Morse pair with

$$(M_{\text{tan}}^+ \times J_{\nu_g}, (M_{\text{tan}}^- \times J_{\nu_g}) \cup (M_{\text{tan}}^+ \times K_{\nu_g})).$$

The sheaf-coefficient form of the stratified Morse product theorem [GM88, Theorem 3.7, pp. 65–66, and §6.A.1, pp. 222–223] applies directly to the constructible complex C . After shrinking U , the controlled product trivialization in that theorem transports C locally constantly in the B -direction. Excision and the relative Künneth isomorphism therefore identify the full local Morse complex with the tensor product of the tangential and normal Morse complexes. The relative cohomology of the tangential Morse pair is a one-dimensional orientation line concentrated in a single degree. Consequently, there is a noncanonical isomorphism in $D^b(\text{Vect}_{\mathbb{C}})$:

$$(3.7) \quad (R\Gamma_{\{f_g \geq 0\}}C)_{x_g} \simeq R\Gamma(J_{\nu_g}, K_{\nu_g}; C)[m] \otimes \mathfrak{o}.$$

for some integer m and some one-dimensional orientation line $\mathfrak{o}$. Shifting and tensoring with the one-dimensional line $\mathfrak{o}$ are conservative, so the right-hand side is nonzero because $\mu_{\nu_g}(C) \neq 0$. The local cohomology complex on the left is therefore nonzero, and [KS85, Theorem 3.1.1] gives $(x_g, 2\text{Re}(e)) \in \text{SS}(C)$. Under the cotangent convention of Section 3.1, this is precisely $\nu_g \in \text{SS}(C)$. $\square$

Lemma 3.4. *Let $C \in \mathcal{D}_g$. Suppose a visible simple perverse sheaf $P_g(\eta)$ has nonzero coefficient in $[C]_{K_0}$. Then $\nu_g \in \text{SS}(C)$.*

Proof. Write $[C]_{K_0} = \sum_j (-1)^j [{}^p\mathcal{H}^j(C)]$. A nonzero total coefficient of $P_g(\eta)$ implies that it is a Jordan–Hölder constituent of ${}^p\mathcal{H}^j(C)$ for at least one j . Exactness of $\chi_{\psi^N}^{\text{mic}}$ gives

$$\dim \text{For } \chi_{\psi^N}^{\text{mic}}({}^p\mathcal{H}^j(C)) = \sum_Q [{}^p\mathcal{H}^j(C) : Q] \dim \text{For } \chi_{\psi^N}^{\text{mic}}(Q).$$

All summands are nonnegative, and the summand indexed by the visible simple $P_g(\eta)$ is positive. Hence $\text{For } \chi_{\psi^N}^{\text{mic}}({}^p\mathcal{H}^j(C)) \neq 0$. By Lemma 3.2, $\mu_{\nu_g}(C) \neq 0$, and Lemma 3.3 gives $\nu_g \in \text{SS}(C)$. $\square$

4. TRANSITION OF SINGULAR SUPPORT

Fix once and for all a nondegenerate ${}^\vee G^\Gamma$ -invariant symmetric complex bilinear form B on ${}^\vee \mathfrak{g}$, as in [AIMV26, Proposition 3.7.1], and use the same symbol for its restriction to ${}^\vee \mathfrak{t}$. This restriction is nondegenerate and ${}^\vee \theta$ -invariant. It identifies ${}^\vee \mathfrak{t}^*$ with ${}^\vee \mathfrak{t}$; in particular, for every parabolic ${}^\vee \mathfrak{p} \subset {}^\vee \mathfrak{t}$ with nilradical ${}^\vee \mathfrak{u}$, one has ${}^\vee \mathfrak{p}^\perp = {}^\vee \mathfrak{u}$. Under the convention of Section 3.1, twice the real part of B realizes the corresponding real-cotangent identification. All identifications of Arthur covectors with the element e , and all cotangent-space identifications in Proposition 4.2, are made using this same form.

All closures of orbits and double cosets in this section are taken in the indicated complex algebraic varieties.

Lemma 4.1. *Let $\pi_f : {}^\vee R \rightarrow X_f = {}^\vee R / {}^\vee P_f$ be the quotient map and $\Phi : {}^\vee R \times {}^\vee P_g X_f \rightarrow X_g \times X_f$ the isomorphism $\Phi([s, y]) = (s {}^\vee P_g, sy)$. Then*

$$\overline{{}^\vee R \cdot (x_g, x_f)} = \Phi({}^\vee R \times {}^\vee P_g \overline{{}^\vee P_g \cdot x_f}), \quad \pi_f^{-1}(\overline{{}^\vee P_g \cdot x_f}) = \overline{{}^\vee P_g {}^\vee P_f}.$$

In particular, the fiber of the first closure over x_g is $\{x_g\} \times \overline{{}^\vee P_g \cdot x_f}$.

Proof. Under Φ , the diagonal orbit corresponds to ${}^\vee R \times {}^\vee P_g ({}^\vee P_g \cdot x_f)$. Since $\overline{{}^\vee P_g \cdot x_f}$ is a closed ${}^\vee P_g$ -stable subvariety of X_f , its associated subbundle is closed in ${}^\vee R \times {}^\vee P_g X_f$; the orbit subbundle is dense in it. This proves the first identity and the assertion about its fiber. For the second, put $A = {}^\vee P_g {}^\vee P_f = \pi_f^{-1}({}^\vee P_g \cdot x_f)$. For $p \in {}^\vee P_f$, right multiplication by p is a homeomorphism and $Ap = A$; hence $\overline{Ap} = \overline{A} = \overline{A}$. Thus $\overline{A}$, and therefore its complement, is right- ${}^\vee P_f$ -saturated. Since π_f is open, that complement is the inverse image of its open image under π_f . The closed set $\pi_f^{-1}(\overline{{}^\vee P_g \cdot x_f})$ contains A , so it contains $\overline{A}$. Conversely, the open image of the complement of $\overline{A}$ is disjoint from ${}^\vee P_g \cdot x_f$, hence from its closure. Taking inverse images and then complements gives $\pi_f^{-1}(\overline{{}^\vee P_g \cdot x_f}) = \overline{A}$. $\square$

Proposition 4.2. *For every $P \in \mathcal{P}_f$, one has $\nu_g \in \text{SS}(IP) \implies \nu_f \in \text{SS}(P)$.*

Proof. Via the normalized equivalences of Lemma 2.2, we can write P also for $\text{Fib}_f(P)$ and write $I = (q_g)_! q_f^*$ on the flag-variety categories. All sheaf operations in this section are derived. We first treat $e = 0$. By [AIMV26, Proposition 3.7.1], the stabilizer orbit of e is open and dense in the conormal fiber. If $e = 0$, that orbit is the singleton $\{0\}$, so the conormal fiber is zero and the corresponding ${}^\vee K$ -orbit in the flag fiber is open. When $e = 0$, Lemma 2.3 gives $\overline{{}^\vee P_f {}^\vee P_g} = {}^\vee P_f$, and hence ${}^\vee P_g \subseteq {}^\vee P_f$. Therefore $Z = X_g$, $q_g = \text{id}$, and $I \simeq q_f^*$. The hypothesis gives $(x_g, 0) \in \text{SS}(q_f^* P)$. The smooth inverse-image formula (3.2) gives $\xi \in \text{SS}(P)_{x_f}$ with ${}^t dq_f \xi = 0$. Since q_f is a submersion, ${}^t dq_f$ is injective. Thus $\xi = 0$, and $(x_f, 0) \in \text{SS}(P)$.

Assume henceforth that $e \neq 0$. Factor $q_g = p_g \circ \iota$, where $\iota : Z \rightarrow X_g \times X_f$ is given by $\iota(z) = (q_g(z), q_f(z))$ and p_g is the first projection. Set $\mathcal{F} = \iota_! q_f^* P$. Then $IP \simeq (p_g)_! \mathcal{F}$. Since X_f is projective, p_g is proper.

Suppose $\nu_g = (x_g, e) \in \text{SS}((p_g)_! \mathcal{F})$. By (3.1), there exists $y \in X_f$ such that

$$(4.1) \quad ((x_g, y), (e, 0)) \in \text{SS}(\mathcal{F}).$$

Write $y = r {}^\vee P_f$.

For a ${}^\vee K$ -orbit $S \subset X_f$ and an ${}^\vee R$ -orbit $\mathcal{T} \subset X_g \times X_f$, the subsets $\mathcal{S}_{S, \mathcal{T}} = (X_g \times S) \cap \mathcal{T}$ form the smooth locally closed stratification used in [AIMV26, (4.6.5)]. In particular,

$$(4.2) \quad \text{SS}(\iota_! q_f^* P) \subseteq \bigcup_{\substack{S \in {}^\vee K \backslash X_f \\ \mathcal{T} \subseteq {}^\vee R \cdot (x_g, x_f)}} T_{\mathcal{S}_{S, \mathcal{T}}}^*(X_g \times X_f).$$

Here $\mathcal{T}$ ranges over the ${}^\vee R$ -orbits contained in the indicated diagonal-orbit closure, and each $T_{\mathcal{S}_{S, \mathcal{T}}}^*(X_g \times X_f)$ is the ordinary conormal bundle of the stratum.

By (4.1) and (4.2), the covector $((x_g, r {}^\vee P_f), (e, 0))$ belongs to one of these conormal bundles. Formula [AIMV26, (4.6.8)], evaluated at this covector, gives $e \in {}^\vee \mathfrak{u}_g \cap \text{Ad}(r) {}^\vee \mathfrak{u}_f$ and $e \in (\text{Ad}(r) {}^\vee \mathfrak{u}_f)^{-\vee \theta}$. In particular,

$$(4.3) \quad e \in \text{Ad}(r) {}^\vee \mathfrak{u}_f.$$

Every base point of a microsupport covector lies in the support of the complex. Hence (4.1) gives $(x_g, r {}^\vee P_f) \in \text{Supp}(\mathcal{F})$. Since $\mathcal{F} = \iota_! q_f^* P$ and $\iota(Z) = {}^\vee R \cdot (x_g, x_f)$, one has $\text{Supp}(\mathcal{F}) \subseteq \overline{{}^\vee R \cdot (x_g, x_f)}$. Thus $(x_g, r {}^\vee P_f) \in \overline{{}^\vee R \cdot (x_g, x_f)}$. Lemma 4.1 first gives $r {}^\vee P_f \in \overline{{}^\vee P_g \cdot x_f}$ and then

$$(4.4) \quad r \in \overline{{}^\vee P_g {}^\vee P_f}.$$

Inversion is an algebraic automorphism of ${}^\vee R$, so (4.3) and (4.4) give $r^{-1} \in \overline{{}^\vee P_f} {}^\vee P_g$ and $\text{Ad}(r^{-1})e \in {}^\vee \mathfrak{u}_f$. By Lemma 2.3, $r^{-1} \in {}^\vee P_f$, and therefore $y = x_f$. We have proved

$$(4.5) \quad ((x_g, x_f), (e, 0)) \in \text{SS}(\iota_1 q_f^* P).$$

The incidence map ι is a locally closed immersion. Choose an open neighborhood $j : V \hookrightarrow X_g \times X_f$ of (x_g, x_f) such that the restriction $\iota_V : U \hookrightarrow V$ is a closed immersion, where $U = \iota^{-1}(V)$. By (3.3), (4.5) remains valid after restriction to V , and $j^{-1}(\iota_1 q_f^* P) \simeq (\iota_V)_*((q_f^* P)|_U)$. Let $z_0 = 1({}^\vee P_g \cap {}^\vee P_f) \in Z$. The equality for a closed immersion in Proposition 3.1(i), followed by locality (3.3), gives

$$(4.6) \quad {}^t d\iota_{z_0}(e, 0) \in \text{SS}(q_f^* P)_{z_0}.$$

Since $\iota = (q_g, q_f)$,

$$(4.7) \quad {}^t d\iota_{z_0}(e, 0) = {}^t dq_g(e).$$

The map q_f is a smooth submersion. By (3.2), (4.6), and (4.7), there exists $\xi \in \text{SS}(P)_{x_f}$ such that

$$(4.8) \quad {}^t dq_f(\xi) = {}^t dq_g(e).$$

We now identify ξ . Use the invariant complex bilinear form fixed above, and twice its real part for the real cotangent bundles. At the base points, $T_{x_a}^* X_a \simeq {}^\vee \mathfrak{p}_a^\perp = {}^\vee \mathfrak{u}_a$ for $a \in \{f, g\}$, while $T_{z_0}^* Z \simeq \text{Lie}({}^\vee P_f \cap {}^\vee P_g)^\perp$. Since $T_{1 \in {}^\vee P_a}({}^\vee R / {}^\vee P_a) = {}^\vee \mathfrak{r} / {}^\vee \mathfrak{p}_a$, the transpose differentials ${}^t dq_f$ and ${}^t dq_g$ are the natural inclusions ${}^\vee \mathfrak{p}_a^\perp \hookrightarrow \text{Lie}({}^\vee P_f \cap {}^\vee P_g)^\perp$.

Write $\lambda_a = \frac{1}{2}h + \zeta_a$. The elements ζ_a centralize the Arthur $\mathfrak{sl}_2$, so $[\lambda_a, e] = \frac{1}{2}[h, e] + [\zeta_a, e] = e$. Hence $e \in {}^\vee \mathfrak{u}_f \cap {}^\vee \mathfrak{u}_g$. Under the inclusions just described,

$$(4.9) \quad {}^t dq_f(e) = {}^t dq_g(e).$$

Combining (4.8) and (4.9) yields ${}^t dq_f(\xi) = {}^t dq_f(e)$. Since q_f is a submersion, ${}^t dq_f$ is injective. Therefore $\xi = e$, and $\nu_f \in \text{SS}(P)$. $\square$

5. COHERENT CONTINUATION OF INDIVIDUAL PACKET MEMBERS

Let $\mathcal{D}(G^\Gamma)$ be the set of G -conjugacy classes of strong real forms. For each $[\gamma] \in \mathcal{D}(G^\Gamma)$, choose a representative, let K_γ be the complexification of a maximal compact subgroup of $G(\mathbb{R}, \gamma)$, and abbreviate $K\Pi_\lambda^z(\mathfrak{g}, K_\gamma) = KM_{\lambda, \text{fl}}^z(\mathfrak{g}, K_\gamma)$. Here $M_{\lambda, \text{fl}}^z(\mathfrak{g}, K_\gamma)$ is the category of finite-length projective type- z $(\mathfrak{g}, K_\gamma)$ -modules with infinitesimal character λ , and $KM_{\lambda, \text{fl}}^z(\mathfrak{g}, K_\gamma)$ is its Grothendieck group. Then we have a natural identification

$$(5.1) \quad K\Pi_\lambda^z(G^\Gamma) \xrightarrow{\sim} \bigoplus_{[\gamma] \in \mathcal{D}(G^\Gamma)} K\Pi_\lambda^z(\mathfrak{g}, K_\gamma).$$

The realization (5.1) depends on the chosen representatives and maximal compact subgroups. Conjugation relates any two choices, and Lemma 5.4 shows that this transport commutes with continuation. Since λ_g is in the good range for L_N and $\lambda_f \in \lambda_g + X^*(L_N)$, each $M \in K\Pi_{\lambda_g}^z(\mathfrak{g}, K_\gamma)$ determines a unique L_N -coherent family $\Theta_{M, \gamma}$ with $\Theta_{M, \gamma}(\lambda_g) = M$; see [AIMV26, Proposition 2.10.2]. On the summand indexed by γ , [AIMV26, Definition 2.10.3] defines coherent continuation by

$$\begin{aligned} T_{L_N, \gamma}(\lambda_g \rightarrow \lambda_f) : K\Pi_{\lambda_g}^z(\mathfrak{g}, K_\gamma) &\longrightarrow K\Pi_{\lambda_f}^z(\mathfrak{g}, K_\gamma), \\ M &\longmapsto \Theta_{M, \gamma}(\lambda_f). \end{aligned}$$

Under (5.1), the global continuation is represented summandwise by

$$(5.2) \quad \bigoplus_{[\gamma] \in \mathcal{D}(G^\Gamma)} T_{L_N, \gamma}(\lambda_g \rightarrow \lambda_f).$$

Thus continuation preserves summands on strong real forms.

Proposition 5.1. *For every $\tau_g \in \Pi^z(G^\Gamma)_{\psi^N}^{\text{mic}}$, $[T\tau_g] \subseteq \Pi^z(G^\Gamma)_{\psi}^{\text{mic}}$.*

Proof. Write $\tau_g = \pi_g(\eta)$, and let $\pi_f(\xi)$ be an irreducible having nonzero coefficient in $T\pi_g(\eta)$. By (2.1), $\langle T\pi_g(\eta), P_f(\xi) \rangle_f \neq 0$. The representation and sheaf Grothendieck groups decompose as orthogonal direct sums over the geometric components indexed by Y . Since convolution is defined for a fixed Y in [AIMV26, Section 3.5], it is block diagonal for these decompositions. Let Y_η be the component containing $P_g(\eta)$. Positive translation fixes the y -coordinate, so the corresponding f - and g -blocks have the common flag-variety model of Lemma 2.2.

If $P_f(\xi)$ belonged to a different Y -block, the right-hand side of (2.9) would vanish. Thus $P_f(\xi)$ belongs to the block corresponding to Y_η .

By (2.9), $0 \neq \langle T\pi_g(\eta), P_f(\xi) \rangle_f = \langle \pi_g(\eta), [IP_f(\xi)]_{K_0} \rangle_g$. The g -side pairing is diagonal, so $P_g(\eta)$ has nonzero coefficient in $[IP_f(\xi)]_{K_0}$. Since $\pi_g(\eta)$ belongs to the translated packet, $P_g(\eta)$ is visible. By Lemma 3.4, $\nu_g \in \text{SS}(IP_f(\xi))$. Proposition 4.2 gives $\nu_f \in \text{SS}(P_f(\xi))$, and the packet criterion (2.2) gives $\pi_f(\xi) \in \Pi^z(G^\Gamma)_\psi^{\text{mic}}$. $\square$

For the argument on coherent families, we must realize the two nonzero strong-real-form summands of extended induction on one compatible chain of module categories. The induction maps themselves are already defined in [AIMV26, (2.6.1)].

Lemma 5.2. *Let G_0 be a real reductive group.*

- (i) *If $P_0 = L_0 N_0$ is a real parabolic with a specified real Levi factor, there is a Cartan involution θ of G_0 such that $\theta(L_0) = L_0$ and $\theta(\mathfrak{p}) = \mathfrak{p}^{\text{op}}$.*
- (ii) *Let σ be the real conjugation on the complexification of G_0 . If $Q = LU$ is a parabolic such that $\sigma(Q) = Q^{\text{op}}$ with common Levi L , there is a Cartan involution θ commuting with σ such that $\theta(L) = L$ and $\theta(Q) = Q$.*

Any two Cartan involutions of a real reductive group are conjugate by an inner automorphism from the identity component.

Proof. For (i), begin with any Cartan involution. The Langlands decomposition of P_0 supplies a θ -stable real Levi factor and an opposite nilradical. Any two real Levi factors of P_0 are conjugate by N_0 , so conjugating θ by an element of N_0 gives the specified L_0 .

For (ii), choose an integral grading element $H_Q \in \mathfrak{z}(\mathfrak{l})$ for Q . Since $\sigma(Q) = Q^{\text{op}}$, the element $-\sigma(H_Q)$ lies in the same open grading chamber as H_Q . The integral element $H_Q - \sigma(H_Q)$ therefore defines the same parabolic and satisfies $\sigma(H_Q - \sigma(H_Q)) = -(H_Q - \sigma(H_Q))$; replace H_Q by this element. Then $iH_Q \in \mathfrak{l}_0$ generates a compact central torus $T_c \subset L_0$. Choose a Cartan involution θ_L of L_0 . Its fixed maximal compact subgroup contains T_c , since T_c is central, so $\theta_L(H_Q) = H_Q$. By [AT18, Theorem 3.13(1)(b)], θ_L extends to a Cartan involution θ of G_0 preserving L_0 . It fixes H_Q and therefore preserves the nonnegative grading parabolic Q . The conjugacy assertion is [AT18, Theorem 3.13(1)(a)]. $\square$

Lemma 5.3. *Let $\delta_2 \in M_2^\Gamma \setminus M_2$ be a strong real form. Assume that its images $\delta_1 \in M_1^\Gamma$ and $\delta_G \in G^\Gamma$ are again strong real forms. Equivalently, $\delta_2^2 \in Z(M_1)$ and $\delta_1^2 \in Z(G)$, where δ_1 and δ_G denote the successive images of δ_2 . There are compatible choices of maximal compact subgroups whose complexifications satisfy $K_{2,\delta_2} = K_{1,\delta_1} \cap M_2$ and $K_{1,\delta_1} = K_{\delta_G} \cap M_1$. On the corresponding Grothendieck groups of projective type- z_2 , type- z_1 , and type- z modules for $(\mathfrak{m}_2, K_{2,\delta_2})$, $(\mathfrak{m}_1, K_{1,\delta_1})$, and $(\mathfrak{g}, K_{\delta_G})$, respectively, the nonzero restrictions of the two maps in [AIMV26, (2.6.1)] are, respectively, the cohomological induction $R_{\mathfrak{q}_2}^{\mathfrak{m}_1}$ and the normalized real parabolic induction $R_{\mathfrak{q}_1}^{\mathfrak{g}}$.*

Proof. The extended Levi inclusions in the Jordan decomposition are $M_2^\Gamma \subset M_1^\Gamma \subset G^\Gamma$; see [AIMV26, Definition 4.1.1(5),(7)]. Since δ_2 is a strong real form, the required finite-order condition is already satisfied. Its image in M_1^Γ is strong exactly when $\delta_2^2 \in Z(M_1)$, and the next image in G^Γ is strong exactly when $\delta_1^2 \in Z(G)$. Formula [AIMV26, (2.6.1)] sends a summand to zero when the corresponding centrality condition fails. Moreover, Q_2 is θ -stable and Q_1 is real in the extended-group sense. Thus [AIMV26, Lemma 2.6.1] gives $\text{Ad}(\delta_1)Q_2 = Q_2^{\text{op}}$ and $\text{Ad}(\delta_G)Q_1 = Q_1$.

Lemma 5.2 allows us to choose the maximal compact subgroups simultaneously as follows. Start with a Cartan involution θ_G^0 of $G(\mathbb{R}, \delta_G)$ adapted to the real parabolic $Q_1(\mathbb{R})$; it preserves its Levi subgroup $M_1(\mathbb{R}, \delta_1)$ and satisfies $\theta_G^0(\mathfrak{q}_1) = \mathfrak{q}_1^{\text{op}}$. Here $\text{Ad}(\delta_1)Q_2 = Q_2^{\text{op}}$ is the extended-group condition. Part (ii) of the lemma supplies a Cartan involution θ_1 of $M_1(\mathbb{R}, \delta_1)$ that preserves Q_2 and M_2 , so $\theta_1(\mathfrak{q}_2) = \mathfrak{q}_2$. Any two Cartan involutions of $M_1(\mathbb{R}, \delta_1)$ are conjugate by $M_1(\mathbb{R}, \delta_1)$. Hence, after replacing θ_G^0 by $\theta_G = \text{Ad}(m) \circ \theta_G^0 \circ \text{Ad}(m^{-1})$ for some $m \in M_1(\mathbb{R}, \delta_1)$, its restriction to M_1 is θ_1 . This replacement still preserves M_1 and remains adapted to Q_1 , since M_1 normalizes Q_1 and Q_1^{op} . If K_{δ_G} is the complexification of the maximal compact subgroup fixed by θ_G , set $K_{1,\delta_1} = K_{\delta_G} \cap M_1$ and $K_{2,\delta_2} = K_{1,\delta_1} \cap M_2$. These choices give the $(\mathfrak{m}_i, K_{i,\delta_i})$ - and $(\mathfrak{g}, K_{\delta_G})$ -module categories. On this fixed chain, the two nonzero summands of [AIMV26, (2.6.1)] are the Euler-characteristic cohomological induction $R_{\mathfrak{q}_2}^{\mathfrak{m}_1}$ for the θ_1 -stable parabolic $\mathfrak{q}_2$, followed by the normalized real parabolic induction $R_{\mathfrak{q}_1}^{\mathfrak{g}}$ for $Q_1(\mathbb{R})$. $\square$

Lemma 5.4. *Let $g \in G$ satisfy $g\delta_G g^{-1} = \gamma$, and put $H^g = gHg^{-1}$ and $L_N^g = gL_N g^{-1}$. In (5.1), choose $K_\gamma = gK_{\delta_G} g^{-1}$. For $a \in \{f, g\}$, define $\lambda_a^g = \lambda_a \circ \text{Ad}(g^{-1}) \in (\text{Lie } H^g)^*$; conjugation identifies $X^*(H)$ with $X^*(H^g)$*

and $X^*(L_N)$ with $X^*(L_N^g)$. Transport of module structures by $\text{Ad}(g)$ gives isomorphisms $C_g : K\Pi_{\lambda_a}^z(\mathfrak{g}, K_{\delta_G}) \rightarrow K\Pi_{\lambda_a}^z(\mathfrak{g}, K_\gamma)$ for $a \in \{f, g\}$. These isomorphisms preserve projective type and satisfy

$$(5.3) \quad C_g \circ T_{L_N, \delta_G}(\lambda_g \rightarrow \lambda_f) = T_{L_N^g, \gamma}(\lambda_g^g \rightarrow \lambda_f^g) \circ C_g.$$

Under the conjugacy identification of [AIMV26, Definition 2.4.2], the operator on the right is the γ -summand denoted by $T_{L_N, \gamma}(\lambda_g \rightarrow \lambda_f)$ in (5.2).

Proof. Since G is connected, choose a path g_t from 1 to g . The maps $x \mapsto g_t x g_t^{-1}$ give a based homotopy from the identity to $\text{Int}(g)$. Equivalently, inner conjugation acts trivially on $\pi_1(G)$. Consequently, on every finite cover in the canonical-cover system, $\text{Int}(g)$ has a unique lift fixing the identity above 1. These lifts are compatible and fix the covering kernels pointwise; passing to the inverse limit gives the canonical identification used in [AIMV26, Definitions 2.4.2–2.4.3]. The corresponding kernel character and the projective type z are unchanged, so each C_g is an isomorphism of the displayed type- z groups. The inner automorphism $\text{Ad}(g)$ fixes $Z(U(\mathfrak{g}))$ pointwise. Moreover, the transport of any finite-dimensional algebraic G -module by $\text{Ad}(g)$ is isomorphic to the original module. Hence C_g preserves the tensor identities defining classical coherent families. It carries a classical family on $\lambda_g + X^*(H)$ to one on $\lambda_g^g + X^*(H^g)$ and carries its restriction on L_N to the corresponding restriction on L_N^g . Good range is invariant under this transport, so the uniqueness in [AIMV26, Proposition 2.10.2] and the definition of continuation give (5.3). Under the conjugacy identification of [AIMV26, Definition 2.4.2], the transported Cartan, Levi, and weights are identified with the original abstract data. Thus the operator on the right of (5.3) is the γ -summand denoted by $T_{L_N, \gamma}(\lambda_g \rightarrow \lambda_f)$ in (5.2). $\square$

Lemma 5.5. Fix a strong real form δ_2 of M_2^Γ , and let $\sigma \in \Pi^{z_2}(M_2^\Gamma)_{\psi_2}^{\text{mic}}$ be carried by the δ_2 -summand. Put $\sigma^N = \sigma \otimes \chi^{2N}$. Then $T(\tilde{R}_N \sigma^N) = \tilde{R}_0 \sigma$ in $K\Pi^z(G^\Gamma)$.

Proof. Choose an irreducible module V representing σ^N on the δ_2 -summand, and write $\sigma^N = (\delta_2, V)$. Tensoring by an algebraic character does not change the strong real form. Let δ_1 and δ_G denote the images of δ_2 in M_1^Γ and G^Γ whenever those images are again strong real forms.

For $\mu \in X^*(M_2)$, let $\mathbb{C}_\mu$ denote the corresponding algebraic character of M_2 . Its pullback to the canonical cover is trivial on the covering kernel, so tensoring by $\mathbb{C}_\mu$ preserves projective type z_2 . We shall extend the resulting family from $\lambda_{2,g} + X^*(M_2)$ to $\lambda_{2,g} + X^*(H)$ as an abstract coherent family.

The inverse character χ^{-2N} determines $\mu_0 = -2N\mu_\chi = -N\kappa \in X^*(M_2) \cap X^*(L_N)$, with $\mathbb{C}_{\mu_0} = \chi^{-2N}$. Regard $X^*(M_2)$ as a sublattice of the common ambient weight lattice $X^*(H)$. The first induction family is prescribed on this sublattice and will then be extended to a classical coherent family on $\lambda_{2,g} + X^*(H)$. Then

$$(5.4) \quad \lambda_{2,g} + \mu_0 = \lambda_{2,f}, \quad \lambda_g + \mu_0 = \lambda_f, \quad \sigma^N \otimes \mathbb{C}_{\mu_0} \simeq \sigma.$$

The first two identities follow from (2.4), and the third from $\sigma^N = \sigma \otimes \chi^{2N}$. Thus both the induction family and T may be evaluated at μ_0 .

The induction on extended groups is defined separately on each summand of strong real form. A summand is sent to zero unless its strong real form remains strong in the larger extended group. Otherwise the two stages are the cohomological and normalized real parabolic induction maps described above. The behavior of coherent families is given by [AIMV26, Proposition 2.10.4 and Proposition 2.8.4]. We work throughout in Grothendieck groups. For $\mu \in X^*(M_2)$, put $v_\mu = [V \otimes \mathbb{C}_\mu]$ and, when $\delta_2^2 \in Z(M_1)$, put $w_\mu = R_{q_2}^{m_1} v_\mu$. Thus w_μ is a Grothendieck class. The first stage sends (δ_2, v_μ) to (δ_1, w_μ) when $\delta_2^2 \in Z(M_1)$ and to zero otherwise. When it is nonzero, the second stage sends (δ_1, w_μ) to $(\delta_G, R_{q_1}^g w_\mu)$ when $\delta_1^2 \in Z(G)$ and to zero otherwise. The choice of branch depends only on δ_2 , not on μ . Since $\tilde{R}_N \sigma^N = \alpha_{\sigma, N} \tau_g \neq 0$ at $\mu = 0$, both centrality conditions hold. The first branch therefore applies for every μ , although an individual induced class may vanish. We work in the compatible module categories of Lemma 5.3; all induced classes lie in the indicated $(\mathfrak{m}_i, K_{i, \delta_i})$ - and $(\mathfrak{g}, K_{\delta_G})$ -module Grothendieck groups. If the representative chosen in (5.1) is γ , we transport the resulting target coherent family at the end using (5.3).

For $\mu \in X^*(M_2)$, put $\Theta_1(\lambda_{2,g} + \mu) = w_\mu = R_{q_2}^{m_1} v_\mu$ in $K\Pi^{z_1}(\mathfrak{m}_1, K_{1, \delta_1})$. The projective types at the first stage are $z_2 \rightarrow z_1$ by (1.1). By [AIMV26, Proposition 2.10.4], Θ_1 is the restriction to the M_2 -character lattice of a classical coherent family for $(\mathfrak{m}_1, K_{1, \delta_1})$. Choose such an extension Θ_1^{cl} of projective type z_1 on $\lambda_{2,g} + X^*(H)$. By [AIMV26, Proposition 2.8.4], pointwise normalized real parabolic induction gives a classical coherent family for $(\mathfrak{g}, K_{\delta_G})$ of projective type z . The type change is $z_1 \rightarrow z$ by (1.1), and the infinitesimal character $\lambda_{2,g} + \mu$ maps to $\lambda_g + \mu$. Denote this family by the formula $\Theta_2^{\text{cl}}(\lambda_g + \mu) = R_{q_1}^g \Theta_1^{\text{cl}}(\lambda_{2,g} + \mu)$ for $\mu \in X^*(H)$. On the

M_2 -character lattice its values are $\Theta_2(\lambda_g + \mu) = R_{q_1}^g \Theta_1(\lambda_{2,g} + \mu)$ in $K\Pi^z(\mathfrak{g}, K_{\delta_G})$. Its values on $X^*(M_2)$ do not depend on the chosen classical extension.

It remains to incorporate Kottwitz signs. On these fixed summands the source and target Kottwitz operators act by the constants $\epsilon_2 = e(M_2(\mathbb{R}, \delta_2))$ and $\epsilon_G = e(G(\mathbb{R}, \delta_G))$, both in $\{\pm 1\}$. Therefore $\tilde{R}(\sigma^N \otimes \mathbb{C}_\mu) = \epsilon_G \epsilon_2 \Theta_2(\lambda_g + \mu)$ for every $\mu \in X^*(M_2)$. For the member σ^N in (2.8), this gives $\alpha_{\sigma,N} = \epsilon_G \epsilon_2 \beta_{\sigma,N}$. Multiplication by $\epsilon_G \epsilon_2$ preserves the coherent-family identities, so $\Theta^{\text{cl}} := \epsilon_G \epsilon_2 \Theta_2^{\text{cl}}$ is a classical coherent family for $(\mathfrak{g}, K_{\delta_G})$ on $\lambda_g + X^*(H)$. For every $\mu \in X^*(M_2)$ it satisfies

$$(5.5) \quad \Theta^{\text{cl}}(\lambda_g + \mu) = \tilde{R}(\sigma^N \otimes \mathbb{C}_\mu).$$

Restricting this classical family to the auxiliary character lattice gives, by [AIMV26, Definition 2.10.1], the L_N -coherent family $\Theta^{L_N} = \Theta^{\text{cl}}|_{\lambda_g + X^*(L_N)}$ in $K\Pi^z(\mathfrak{g}, K_{\delta_G})$. It is viewed through (5.1) as the fixed δ_G -summand of $K\Pi^z(G^\Gamma)$.

Since $0, \mu_0 \in X^*(L_N)$ and $\mu_0 \in X^*(M_2)$, equations (5.4) and (5.5) give $\Theta^{L_N}(\lambda_g) = \tilde{R}_N \sigma^N$ and $\Theta^{L_N}(\lambda_f) = \tilde{R}_0 \sigma$. By Lemma 2.1(1), the parabolic subalgebra $\mathfrak{q}_N = \mathfrak{l}_N \oplus \mathfrak{u}_N$ has Levi factor $\mathfrak{l}_N$, and λ_g is in the good range for $\mathfrak{q}_N$. Thus λ_g is in the good range for $\mathfrak{l}_N$ in the sense used in [AIMV26, Definition 2.5.3]. By [AIMV26, Proposition 2.10.2], applied in the $(\mathfrak{g}, K_{\delta_G})$ -module category, an L_N -coherent family is uniquely determined by its value at λ_g . Thus Θ^{L_N} is independent of the chosen classical extension, and [AIMV26, Definition 2.10.3] gives $T_{L_N, \delta_G}(\lambda_g \rightarrow \lambda_f) \Theta^{L_N}(\lambda_g) = \Theta^{L_N}(\lambda_f)$. Hence $T_{L_N, \delta_G}(\lambda_g \rightarrow \lambda_f) \tilde{R}_N \sigma^N = \tilde{R}_0 \sigma$. If the representative in (5.1) is γ , choose g as above and apply C_g . Conjugacy naturality (5.3) identifies the transported continuation with the γ -summand, and [AIMV26, Definition 2.4.2] identifies the transported endpoint classes with the same elements of $K\Pi^z(G^\Gamma)$. Thus (5.2) gives the asserted identity in $K\Pi^z(G^\Gamma)$. $\square$

6. PROOF OF THE SUPPORT EQUALITY

Theorem 6.1. *One has*

$$(6.1) \quad \left[R_{q_1}^g R_{q_2}^{m_1} \Pi^{zz(\rho_{u_1})z(\rho_{u_2})}(M_2^\Gamma)_{\psi_2}^{\text{mic}} \right] \subseteq \Pi^z(G^\Gamma)_{\psi}^{\text{mic}}.$$

Proof. Fix $\sigma \in \Pi^{z_2}(M_2^\Gamma)_{\psi_2}^{\text{mic}}$. By (2.3), $\sigma^N \in \Pi^{z_2}(M_2^\Gamma)_{\psi_2^N}^{\text{mic}}$. The good-range statement (2.8) gives a unique $\tau_g \in \Pi^z(G^\Gamma)_{\psi_N}^{\text{mic}}$ and a sign $\alpha_{\sigma,N}$ such that $\tilde{R}_N \sigma^N = \alpha_{\sigma,N} \tau_g$. Applying T and Lemma 5.5 gives $\tilde{R}_0 \sigma = T(\tilde{R}_N \sigma^N) = \alpha_{\sigma,N} T \tau_g$. By Proposition 5.1, $[T \tau_g] \subseteq \Pi^z(G^\Gamma)_{\psi}^{\text{mic}}$. Hence $[\tilde{R}_0 \sigma] \subseteq \Pi^z(G^\Gamma)_{\psi}^{\text{mic}}$. Since σ is irreducible, it belongs to a single strong-real-form summand. Write $e(M_2^\Gamma) \sigma = \epsilon_2 \sigma$, where $\epsilon_2 \in \{\pm 1\}$. From $\tilde{R}_0 = e(G^\Gamma) R_0 e(M_2^\Gamma)$ we obtain $R_0 \sigma = \epsilon_2 e(G^\Gamma) \tilde{R}_0 \sigma$. The operator $e(G^\Gamma)$ is diagonal with entries ± 1 in the irreducible basis, and hence $[R_0 \sigma] = [\tilde{R}_0 \sigma]$. By definition, $R_0 \sigma$ is $R_{q_1}^g R_{q_2}^{m_1} \sigma$. Taking the union over all source-packet members proves (6.1). $\square$

The opposite inclusion is (1.2), proved in [AIMV26, Theorem 4.6.1(iii)].

Proof of Theorem 1.1. Combine Theorem 6.1 with (1.2). $\square$

REFERENCES

- [ABV92] J. Adams, D. Barbasch, and D. Vogan, *The Langlands Classification and Irreducible Characters for Real Reductive Groups*, Progress in Mathematics, vol. 104, Birkhäuser, Boston, 1992. **2**, **5**, **6**
- [AT18] J. Adams and O. Taïbi, *Galois and Cartan Cohomology of Real Groups*, Duke Math. J. **167** (2018), no. 6, 1057–1097. **9**
- [AIMV26] J. Adams, A. Ionov, L. Mason-Brown, and D. Vogan, *The Unitarity of Arthur Packets for Real Reductive Groups*, arXiv:2606.01609v1, June 1, 2026. **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**, **9**, **10**, **11**
- [GM88] M. Goresky and R. MacPherson, *Stratified Morse Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 14, Springer-Verlag, Berlin, 1988. **6**
- [KS85] M. Kashiwara and P. Schapira, *Microlocal Study of Sheaves*, Astérisque **128** (1985), 1–235. **5**, **6**

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